

AYUSH SHRESTHA

Software Developer · Warehouse Automation & Enterprise Systems

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SUMMARY

Software Developer with 5+ years enhancing and supporting warehouse automation software at E80 Group. Works across the full stack — C#/.NET application features, SQL Server performance, ERP integrations, and production troubleshooting — in a live industrial environment where LGVs, palletizers, and conveyors run 24/7. Experienced integrating with ERP systems like SAP, JD Edwards, Manhattan, PrIME etc. via REST/XML, configuring SQL Server high availability, and working alongside operations teams to deliver reliable system behavior.

TECHNICAL SKILLS

Languages: C#, Python, Java, C++, SQL, JavaScript, HTML/CSS, PHP

Frameworks & Patterns: .NET (MVC, WCF, Web API, Windows Services), React, Node.js, Bootstrap, OSGi, Apache Sling

Databases: SQL Server (stored procedures, index/fragmentation, Failover Clustering, Always On AG), Oracle, NoSQL/Firebase

Infrastructure: Docker, MSMQ, Microsoft DCOM, Azure, AWS (Lambda, EC2), Azure DevOps, Jenkins CI/CD

Industrial / WMS: OPC UA / OPC Server, PLC integration, LGV fleet management, WMS ↔ ERP (SAP, JD Edwards), voice picking

Tools: Postman, Git, CVS, Windchill, MDM

EXPERIENCE

Software Developer · E80 Group

Chicago, IL | May 2021 – Present

E80 Group builds autonomous mobile robots (LGVs) and warehouse automation software for manufacturing and distribution clients. Works on the C#/.NET WMS platform — implementing features, maintaining integrations, and supporting production operations.

- ▶ Developed and updated C#/.NET WMS features across the full stack — from application logic and MVC architecture to Web API endpoints used for data transfer to third-party systems and client GUIs.
- ▶ Integrated WMS and LGV logical communication with PLCs through an OPC server, enabling real-time coordination between software and physical warehouse equipment.
- ▶ Integrated the WMS with ERP systems like SAP, JD Edwards, Manhattan, PrIME etc. via HTTPS and basic authentication: outbound requests generate XML payloads to create and process direction for different processes, with response parsing to confirm expected system state.
- ▶ Worked end-to-end on an ERP-to-voice-picking workflow — from shipment creation in the E80 system through to case-pick execution in a voice picking application, coordinating data handoffs across systems.
- ▶ Led software changes to accommodate a fleet-wide battery transition from lead-acid (battery-swap charging) to lithium-ion (inductive charging) — validated LGV configurations behaved correctly under the new charging model and resolved edge cases in dispatch logic.
- ▶ Modified active locations like inbound and production line location behavior in the WMS so LGVs make correct pick/drop and routing decisions based on updated business rules; integrated operational alarms into the UI for real-time equipment visibility.
- ▶ Deployed and maintained Windows Services on Azure and AWS for high-availability, high-performance operation; used WCF to meet client GUI integration requirements.
- ▶ Worked with Microsoft DCOM for distributed system communication between WMS components and remote services.
- ▶ Worked with SQL Server Failover Clustering and Always On Availability Groups for database and application high availability.
- ▶ Diagnosed and rewrote a recurring SQL maintenance job — reduced execution time from 15 minutes to 3 minutes through query restructuring and index/fragmentation analysis.
- ▶ Used Docker development environments
- ▶ Used MSMQ for inter-service messaging within the WMS platform

- Tested and validated endpoint URLs integrations using Postman.
- Debugged and resolved production incidents across the WMS, LGV fleet, and ERP integrations; worked directly with operations teams to minimize impact during active warehouse shifts.

Software Developer Intern · Comau

Auburn Hills, MI | Aug 2020 – Dec 2020

Comau (Stellantis subsidiary) builds industrial robotics systems. Joined a four-person software team upgrading the ROS control stack for 6 axis e.DO robot.

- Migrated robotics control packages from ROS1 to ROS2 in C++14, a prerequisite for integrating a new ML-based motion planning module.
- Implemented a ROS1 Bridge layer to maintain backward compatibility with production robot messaging during the phased rollout, preventing line downtime during migration.
- Wrote requirements, design specs, and integration test plans for each migrated package, establishing a documentation baseline adopted by the broader team.

IT Assistant · Wayne State University, Science Hall

Detroit, MI | Oct 2019 – Dec 2020

- Resolved hardware, software, and networking issues for 200+ faculty and staff users.
- Managed Active Directory accounts and group policies; ensured secure network access for research computing resources.

EDUCATION

Master of Business Administration (MBA), University of the People

2023

Bachelor of Science in Computer Science, Wayne State University

2020

PROJECTS

Smartphone Application (Capstone) — Led Agile-based full-stack development using JSP, Servlets, JavaScript, HTML, and XML. Authored JUnit test suite covering core business logic.

Personal Portfolio Website — Designed and built responsive portfolio at iamayushshrestha.github.io using HTML/CSS/JavaScript.

CERTIFICATIONS & VOLUNTEERING

Fundamentals of Digital Marketing Certificate · Google

2019

Volunteer Java Tutor · Collider Altimetrik

2020